

Variability in the Effectiveness of a Video Modeling Intervention Package for Children With Autism

Journal of Positive Behavior Interventions
2015, Vol. 17(2) 105–115
© Hammill Institute on Disabilities 2014
Reprints and permissions:
sagepub.com/journalsPermissions.nav
DOI: 10.1177/1098300714548798
jpbj.sagepub.com



Joshua B. Plavnick, PhD¹, Mari C. MacFarland, MA¹,
and Summer J. Ferreri, PhD¹

Abstract

Video modeling is an evidence-based instructional strategy for teaching a variety of skills to individuals with autism. Despite the effectiveness of this strategy, there is some uncertainty regarding the conditions under which video modeling is likely to be effective. The present investigation examined the differential effectiveness of video modeling for teaching peer-directed initiations to three young children with autism using a single subject reversal design. In one condition, sharing toys, participants were taught to initiate with peers in the form of inviting a peer to join an activity that was preferred for the participant. In the second condition, joining a play in progress, participants were taught to ask if they could join in a preferred activity with peers. Participants demonstrated initiations during the joining but not during the sharing conditions. The results suggest video modeling can be differentially effective as a function of the target behavior and corresponding environmental events that are both observed and experienced by the child. These outcomes may have implications for how practitioners select social targets and develop video modeling materials when teaching individuals with autism.

Keywords

autism, initiations, social skills, video modeling

Video modeling is an evidence-based practice for teaching a variety of skills to children with autism and involves showing a target student a video display of another individual performing a specific behavior and then providing an opportunity for the target student to perform the same behavior (Bellini & Akullian, 2007). Developed from early work by Miller and Dollard (1941) in observational learning and later from Bandura's (1977) theory of social learning, video modeling is based on the tenet that individuals can learn new skills through observation of others performing a similar behavior. Although attending to the relevant aspects of a model in natural settings is typically a skill deficit for children with autism, a preference for attending to video may mitigate this deficit and facilitate acquisition of behaviors via video modeling (Cardon & Azuma, 2012). Skills taught to children with autism using video modeling include conversational speech (Charlop & Milstein, 1989), play-related statements (Taylor, Levin, & Jasper, 1999), perspective taking (Charlop-Christy & Daneshvar, 2003), affective responding (Gena, Couloura, & Kymissis, 2005), reciprocal pretend play (MacDonald, Sacramone, Mansfield, Wiltz, & Ahearn, 2009), and social initiations (Nikopoulos & Keenan, 2003, 2007; Tetreault & Lerman, 2010).

Despite a majority of positive outcomes, there is some variability in the effectiveness of video modeling,

the sources of which have yet to be empirically identified (Mason, Ganz, Parker, Burke, & Camargo, 2012). Given the rapid rate of improvement that has been observed when video modeling is effective, research that elucidates the sources of variability could assist practitioners in deciding when and how to use the procedure. Examples of variability observed in the outcomes of previous video modeling research studies are described and analyzed below, with an emphasis on the plausibility of the rationales for hypothesized sources of the variability.

Across Participant Variability

Most variability of video modeling outcomes has been discussed within the context of the intervention working for some individuals with autism, but not others (Mason et al.,

¹Michigan State University, East Lansing, USA

Corresponding Author:

Joshua B. Plavnick, Department of Counseling, Educational Psychology, and Special Education, Michigan State University, 620 Farm Lane, 341 Erickson Hall, East Lansing, MI 48824, USA.
Email: plavnick@msu.edu

Action Editor: V. Mark Durand

2012; Nikopoulos & Keenan, 2003; Sherer et al., 2001). For example, Sherer and colleagues (2001) discovered variability across participants when video modeling was used to teach conversational speech to children with autism. Although three participants acquired conversational skills, two participants did not meet the criterion for acquisition over several months of exposure to the treatment conditions. Similarly, Nikopoulos and Keenan (2003) found that four of seven children with autism improved in social initiations and toy play following exposure to video modeling; the remaining three participants showed no improvement.

It is not clear why video modeling has been effective for some children and not for others. Some posit that there are individual characteristics that increase the likelihood of the intervention working for one child and not another (Delano, 2007). This may be the case when certain prerequisite skills such as attending to the video are not yet mastered, but in other cases, such an explanation may oversimplify the sources of variability and does not account for likely interactions between procedural and participant characteristics (Mason et al., 2012). It is possible, for example, that a child who does not learn from video modeling given a certain set of procedural components could learn from video modeling given a different arrangement of procedural components (see Plavnick & Ferreri, 2011, for an example).

Within Participant Variability

Although less often discussed in previous research on video modeling, instances of within-participant variability clarify the conditions under which the intervention may or may not be effective. All child-specific characteristics are held constant in these situations, which reduces uncertainty about the source of variability. We would presume then that any observed variability is more likely attributable to intentional or unintentional alterations to the intervention procedures.

Tetreault and Lerman (2010) found inconsistent effects when teaching initiations to children with autism using point-of-view video modeling, which depicts the scene from the perspective the target participant will see when he or she performs the target response. In addition to across participant variability in outcomes, one participant did not perform the first skill after viewing a video model, but did perform the other two behaviors taught during the study. Interestingly, the participant learned to ask a social partner for assistance and show a drawing to a social partner, but did not learn to share a toy. Although the difficulty level of these skills was relatively similar, the observed and direct consequences for such behaviors are quite different. When viewing the sharing a toy video, the participant observed the model give the toy away as opposed to observing the model receive assistance or attention for the other behaviors.

Plavnick and Ferreri (2011) observed similar within-participant variability when using video modeling to teach basic requests. The researchers found that children with autism acquired and performed vocal initiations when shown a video model of a child requesting items known to be highly preferred for the participant but did not imitate a video when the consequences for the model were nonpreferred for the participant. In both conditions, the consequences for the participant matched the consequences for the model, which involved contingent delivery of the item requested or adult attention. Importantly, one participant imitated the model for initial trials when the behavior was followed by a nonpreferred consequence, but responding under this condition quickly declined to zero. The video may have had initial stimulus control over the imitative behavior, but responding quickly decreased as it was not directly reinforced. In essence, participants in both studies (i.e., Plavnick & Ferreri, 2011; Tetreault & Lerman, 2010) imitated the behaviors followed by consequences known to be preferred by participants but did not imitate behaviors followed by consequences that were neutral or nonpreferred.

Motivation and Video Modeling

The pattern of variable responding displayed by the participants in the Tetreault and Lerman (2010) and Plavnick and Ferreri (2011) studies aligns with Bandura's (1977) theory about the role of motivation in observational learning. Bandura suggested individuals would be more likely to follow successful behavior as opposed to unsuccessful behavior; successful meaning behaviors followed by observed preferred consequences and unsuccessful meaning behaviors followed by neutral or nonpreferred observed consequences, such as temporarily giving away a preferred toy.

It is important to also consider the direct consequences children with autism experience when video modeling is used to teach new skills as these consequences influence future probability of learning via video modeling. Similar to other behavioral interventions, video modeling procedures can target behaviors that lead to naturally occurring reinforcers or the delivery of unrelated reinforcers contingent on the correct response (Bellini & Akullian, 2007). In studies where the target response contacted a naturally occurring reinforcer (e.g., asking for assistance), participants' motivation to perform the response and continue to participate in the instructional process likely increased (R. L. Koegel & Egel, 1979; L. K. Koegel, Singh, & Koegel, 2010; McGee, Almeida, Sulzer-Azaroff, & Feldman, 1992). However, when the target response did not contact reinforcers in the natural environment (e.g., sharing a toy), motivation to continue engaging in the response likely decreased.

Video modeling interventions require the same consideration of motivative variables involved in the majority of instructional protocols for children with autism. However,

the environmental events observed in the video introduce additional motivative or discriminative variables that subsequently lead to differences in imitation of a model for an individual. That is, absent a previous history of direct reinforcement for a specific response, a child may be motivated to imitate a video model who obtains a preferred consequence but not motivated to imitate a model who obtains a neutral or aversive consequence (Bandura, 1965). Despite a long history in behavioral research on instructional methods, motivational variables, including naturally occurring reinforcers, have been minimally examined in video modeling research.

The purpose of the present investigation was to experimentally evaluate a source of variability related to motivation and naturally occurring reinforcers when video modeling is used to teach social interaction to children with autism. Specifically, we sought to answer the following research question:

Research Question 1: Does manipulation of the target behavior, and the corresponding environmental events related to that behavior, lead to within-participant variability in video modeling outcomes for young children with autism?

Similar to the procedures used by Plavnick and Ferreri (2011), the response requirement in the present investigation involved equivalent effort across experimental conditions but altered antecedents and consequences associated with the responses. The results contribute to the limited literature demonstrating empirical assessments of outcome variability when video modeling is used as an instructional procedure for children with autism.

Method

Participants and Setting

The experimenters obtained approval from the University Institutional Review Board and then recruited preschool-aged participants who had been diagnosed with autistic disorder and demonstrated minimal social interaction with peers. The first three children who met the inclusion criteria were accepted for participation in the study. The three male participants received special education services under the primary eligibility category of Autism Spectrum Disorders. The classroom teacher reported that all participants demonstrated minimal interaction with peers, only doing so when prompted by adults. This report was confirmed during an informal observation conducted by the researchers. Parents of all participants signed a consent form for their child to participate prior to initiation of research activities. All participants demonstrated some spoken language, but in almost all cases, language consisted of spontaneous four- to five-word echolalic

phrases or fully prompted vocal responses to questions. All participants also attended to a video for at least 15 s and imitated actions (e.g., gestures), actions with objects, and spoken phrases (e.g., “the block is green”) that had been directly taught, but accurate imitation of novel actions and phrases was variable for all participants. That is, they demonstrated instances of generalized imitation, but did not have a strong generalized imitative repertoire.

Vito was 5 years old at the beginning of the research study and was diagnosed with autistic disorder by a licensed psychologist at 34 months. His parents reported typical development until the age of 2 followed by a regression in language, which led to a referral for diagnostic services. When the present investigation began, a language sample collected by Vito’s speech and language pathologist (SLP) indicated his expressive vocabulary consisted of 30 different spoken words typically emitted as requests (e.g., “cracker please”) or prompted answers to questions (e.g., “my name is Vito”). Vito’s score on the *Childhood Autism Rating Scale–2nd Edition* (CARS-2; Schopler, Van Bourgondien, Wellman, & Love, 2010) was a 46.5, which according to the CARS-2 scoring system, places him in the “severely autistic” category. Based on an observational assessment administered prior to the research study, Vito engaged in repetitive motor behavior with toys, left an area if another child entered, and did not initiate interactions with peers. He did attend to adult instructions and initiated interactions with adults.

Reese was 6 years old at the beginning of the research study and had been diagnosed with autistic disorder by an early intervention team that included a licensed psychologist and using criteria established within the *Diagnostic and Statistical Manual of Mental Disorders* (4th ed., text rev.; DSM-IV-TR; American Psychiatric Association, 2000) at the age of 3. Reese’s parents reported that he emitted some words between 12 and 18 months of age but stopped talking by the age of 2, at which time he was diagnosed with apraxia and started receiving speech and language services. Reese was observed by teachers and researchers to emit a variety of two- to three-word requests and imitated one-word responses when prompted by adults when the study started. Reese did not initiate interactions with peers and only took turns with toys or when playing games if prompted by an adult. Reese did approach adults to initiate interactions.

Ivan was 5 years old at the beginning of the study and was diagnosed with autistic disorder at 4 years of age. His score on the *Gilliam Autism Rating Scale–2nd Edition* (Gilliam, 2006) was 106, which indicates an average probability of having autism. Although nonverbal until the age of 3, a vocal language sample collected by the SLP indicated Ivan emitted approximately 25 different words when the study started and could emit four- to five-word requests such as “I want the crayon please.” These requests always followed the same order and format with only the label of

the item varying across requests, suggesting a memorized utterance. Ivan did not initiate interactions with peers or respond to peers when they initiated interactions. He engaged in limited parallel play, imitated adult actions, and initiated interactions with adults.

The research was conducted within a public early childhood center located in a midwestern suburb in the United States. All research activities were administered as part of a social skills group for preschool children with autism and within a 10 × 5 m space, partitioned with bookshelves and cabinets, within a classroom for children with autism. The interventionist was a SLP who oversaw educational programming for the participants and was trained by the first author to implement the social skills group procedures. The SLP, the three student participants (listed above), and sometimes a secondary observer were the only individuals involved in the social skills group during most of the research sessions. However, on some occasions, Reese or Ivan was absent from school. To ensure a total of three children within every social skills group session, another child from the classroom was brought into the social skills group as a replacement for a student who was absent. All other children in the classroom were working in one-to-one arrangements with instructional staff on the other side of the room and were not in any way involved in the social skills group.

Measurement

The dependent variable was initiations directed toward peers. An initiation was defined as walking up to and stopping approximately 1 to 2 m from a peer, emitting an attention gaining response (e.g., “hey” or peer’s name), and then inviting the peer to play with a common toy (i.e., sharing) or asking to join the peer in play (i.e., joining) depending on the experimental conditions (discussed below). Examples of initiations include a student walking up to a peer and stopping approximately 1 m away, then saying “Reese, let’s play” or “Hi, can I play?” during the sharing and joining conditions, respectively. A nonexample of an initiation would be if the student performed a similar physical response as above, emitted the vocalization “hi” and then reached out and grabbed a toy from his peer.

The experimenters recorded the percentage of trials participants emitted the target behavior across all conditions. Specifically, observers recorded an occurrence or nonoccurrence of initiations after the facilitator instructed participants to “ask a peer to play” or “see if you can play with your friends.” The instruction was delivered following depiction of the video in the intervention conditions and given without showing the video during the baseline condition. The total correct responses were divided by the total opportunities to obtain a percentage of correct responding. Three to six opportunities to perform the social initiations were provided for each participant across all sessions.

The primary data collectors were the interventionist and a doctoral student in special education, both of whom were trained by the first author to record instances of the target behaviors during practice sessions occurring prior to the investigation. The first author was a secondary observer and independently recorded dependent measures across 31% of randomly selected sessions evenly distributed across conditions. Interobserver agreement (IOA) was calculated using point-by-point agreement; agreements or disagreements were scored for each trial and total agreements were divided by total agreements plus disagreements and multiplied by 100 to obtain a percentage. Mean IOA was 100% during baseline, 95% (range = 86%–100%) during the sharing condition, and 95% (range = 80%–100%) during the joining condition.

Materials

Several items were used to encourage social interaction during the social skills group sessions. Children’s books, building blocks, and games such as bingo were used during all conditions. In addition, several highly preferred technological devices including small musical keyboards, an Apple iPod Touch® and iPad®, and a toy computer were used during all conditions. Preferred items were identified by asking teachers to complete a preference inventory and then confirmed using a single stimulus approach-based assessment (e.g., Pace, Ivancic, Edwards, Iwata, & Page, 1985). Specifically, all items reported as preferred for multiple participants were then tested using the single stimulus preference assessment procedure (Pace et al., 1985). If all participants accepted and interacted with an item on the first three trials it was presented, the item was identified as preferred and used accordingly during all experimental conditions and for the creation of video clips. A Sony Camcorder® (model-HDR-XR160) and tripod were used to record video clips, which were displayed on an Apple iPad2® during the video modeling conditions.

Video clips. Six video clips, three variations for each condition, were developed to provide models for performing target social skills. Each video was 20- to 30-s long and depicted two typically developing females, ages 3 and 5 years old, engaging in the target response. An important distinction was made between the video clips for each of the video modeling conditions. The videos used during the sharing condition depicted a model approaching a peer while holding an item known to be preferred by participants (e.g., toy computer) and emitting a vocal invitation for the peer to join the model. The peer agreed and the model immediately shared the item with the peer while the two interacted with the item together. Conversely, the videos used during the joining condition depicted a model who did not have a preferred item in her possession. She approached

Table 1. Programmed Antecedents and Consequences for Correct and Incorrect Responses.

Condition	Antecedent(s)	Consequence for correct response	Consequence for incorrect response
Baseline	Teacher instruction to invite peer to play or ask to join peer(s)	Interact with item together with peer	Blocked from access to item and told how to perform correctly the next time
Sharing	Facilitator shows video of model inviting another child to play, then instructs participant to "invite your friend to play." Participant has possession of preferred item	Participant maintains partial access to preferred item while playing with peer	Participant is blocked from any access to item and told how to perform correctly the next time
Joining	Facilitator shows video of model asking to join another peer who is playing with a preferred item, then instructs participant to "see if you can play with a friend." Peer has possession of preferred item	Participant obtains partial access to preferred item and plays with peer	Participant is redirected and told how to perform correctly the next time

a peer who had the preferred item and asked if she could play. The peer agreed and immediately provided the model with a turn to use the preferred item.

Multiple variations of videos were made to promote generalization by incorporating a range of materials familiar to participants and varying the modeled speech to provide multiple examples of how one could engage in the functional aspects of the social skill. Once created, the videos were streamed directly onto an Apple MacBook Pro® and edited using iMovie. The completed video clips were stored on the iPad2® and accessed by the interventionist as needed.

Experimental Design

A single-case reversal design was used to assess for differences between the two video modeling conditions, sharing and joining, on the performance of social initiations. The reversal design in the present investigation included three manipulations of experimental conditions by the researchers. Following an initial baseline condition (A) to assess levels of the targeted initiations without explicit instruction, the experimenters then alternated between sharing (B) and joining (C) conditions (i.e., A–B–C–B–C). The sharing condition involved the assessment of target behaviors after participants observed the sharing video described above and were presented with an opportunity to perform the target behaviors under similar conditions to the sharing video; the participant had a preferred item and was instructed to ask a peer to play. The joining condition involved the assessment of target behaviors when the participants observed the joining video described above and were then presented with an opportunity to perform the target behaviors under similar conditions to the joining video; a peer had a preferred item and the participant was instructed to ask his

peer if he could play. Although the facilitator of the groups knew the procedures associated with each condition, she was naïve to the hypothesis that the condition with a preferred observable consequence would influence behavior.

Procedures

During all conditions, instruction was delivered as part of a social skills group developed by the first and second authors for preschool children with autism. Participants attended two 15-min social skills group sessions each week. One facilitator, the SLP, provided instruction to all participants at one time during each session. To start a session, participants were directed to sit on the floor in a semi-circle facing the instructor within a 4 × 7 m space in the classroom. Procedures then varied based on whether the session was implemented during baseline, sharing, or joining conditions.

Table 1 outlines the specific antecedents and consequences associated with both correct and incorrect responses in each condition. The primary distinction between baseline and the video modeling conditions was whether a video was shown prior to the instruction to perform the target response. The primary distinction between the two video modeling conditions was whether the participant started with access to the preferred item and *maintained* partial access contingent on a correct response (i.e., sharing) or the peer started with access to the item and the participant *obtained* partial access contingent on a correct response (i.e., joining).

Baseline. During baseline sessions, three trials for each target behavior (for a total of six trials per session) were randomly administered to each participant and the experimenters

assessed for social initiations when participants were given an instruction to invite a peer to play together with a preferred item (i.e., sharing) or join a peer who was already playing with a preferred item (i.e., joining). One purpose of the baseline condition was to assess whether the children approached one another when instructed and, if so, whether they performed any aspect of the target behavior prior to instruction. After instructing the students to sit on the floor and securing attention, the facilitator provided an instruction to one student to engage in an activity such as playing with a toy or looking at a book. The facilitator then directed the remaining students to engage in another activity. The activities varied depending on which student was performing a trial and whether the trial was a probe for sharing or joining. When a student was involved in a sharing trial, he was given a high-preference item and instructed to ask his friends to play with him. When a student was involved in a joining trial, his peers were given the high-preference item or activity and he was instructed to ask to join his peers in play.

After setting up a trial and to begin a performance opportunity, the facilitator provided an instruction to the target participant (e.g., “see if Vito wants to play with you”) and then paused to observe the participant’s response. If the participant initiated with peers, he was able to access the preferred item with his peer. If the participant did not emit both the physical and the vocal behaviors associated with an initiation, the facilitator gently blocked him from accessing the preferred consequence and provided a vocal correction by telling the participant what he should do different next time (e.g., “Nice try. Next time, remember to ask your friend to play”). The participant was then instructed to interact with a neutrally preferred stimulus.

Video modeling: General. During both video modeling conditions the facilitator held an iPad within 0.75 m of the participants who sat in a semi-circle around the facilitator. The facilitator instructed the children to watch the video, showed the video clip depicting the target behavior, and then administered a probe for each participant to engage in the target behavior with one of his peers in the social skills group as a social partner. There were no programmed prompts or consequences to promote attending to the video. Periodic probes of attending to video behavior suggested that children oriented toward the video and appeared to watch the clips.

Following the first video display (i.e., two showings of the clip), each participant had one opportunity to perform the target behavior. If successful, participants were given 1 to 2 min to engage in the activity together, depending on the activity associated with that trial. The facilitator then instructed the participants to return to the viewing area, showed a second video model of the same behavior to all participants, and provided a second opportunity for each participant to perform the response. That is, the video depicted the same behavior, but had slight variations in

language and stimuli. This sequence was repeated a third time and, when time allowed, fourth time so that all participants had three to four opportunities to perform the target response during each group instructional session. Participants willingly returned to the viewing area when instructed to watch the video across all trials, sessions, and conditions. As three versions of each behavior had been captured on video, a different version (i.e., clip) was shown for each trial, with the first version repeated if four trials occurred within a session.

The nature of the social skills group required the participants to interact with one another as social partners. Thus, the participant who received the first trial had an opportunity to perform the target behavior immediately following the video display. However, the other two participants had a slight delay prior to performing the target response. To provide each participant with a similar number of immediate and delayed performance opportunities, performance order was counterbalanced across participants for all behaviors, sessions, and conditions.

Sharing. The sharing condition involved simultaneously showing all participants a video of a child inviting a friend to play together with materials that were highly preferred for the participants. The facilitator showed the video clip 2 times and then provided an opportunity for each participant to perform the target behavior by giving the participant a preferred item and instructing him to “see if your friend wants to play with you.”

Differential consequences were administered in the same manner described during the baseline condition; correct responses were followed by prompts as needed to ensure the target participant shared the item or activity with a peer as modeled in the video. The participants could then interact together with the preferred item. Importantly, the participant did not have to give the item completely over to a peer to share the item, but rather had to provide the peer with some access to the item. For example, if the participant was holding the musical keyboard, he had to sit down next to his peer and provide access to a side of the keyboard during the sharing condition for a correct response to be scored. The participant and the peer could then play with the keyboard together, thus allowing some access to the participant for accurate sharing.

If an incorrect response occurred, the facilitator gently removed the preferred item and provided vocal feedback to the target participant, as in baseline. Although the participant started with possession of the item after observing the video, any attempt to play with, manipulate, or otherwise interact with the item prior to inviting his peer to play together with him was blocked. The sequence of showing a video, creating an opportunity to respond, and administering differential consequences was repeated 3 times per session.

Joining. During the joining condition, the video model was displayed as described during the sharing condition though the modeled response was asking to join others as opposed to inviting others to play. When asking to join others, the child in the video approached a peer who had possession of an item known to be preferred for participants and asked to play with that peer. This condition is distinct from the sharing condition because participants observed the model, who did not previously have a preferred item, obtain access to a preferred item or activity with a peer (as opposed to sharing a preferred item with a peer).

After viewing the video, the participants were given an opportunity to imitate the model in a manner similar to that described during the sharing condition. However, during the joining condition, peers were given the preferred stimuli, and the target participant was instructed to “see if you can play with your friends.” Contingent on an accurate response, the target child was able to join his peers, who were prompted as necessary to provide the joining participant with an immediate opportunity to interact with the preferred item or activity. As in the example described during the sharing condition, a participant was required to sit down next to his peer and interact with the item in the same general space as his peer. If the children did not immediately interact with the item together, they were prompted by the facilitator to do so; the response was still scored as correct in this case and participants could still therefore access the item together. All aspects of the condition, such as showing the video, antecedent instructions, order of performance opportunity, and differential consequences, were identical to those reported under the sharing condition.

Results

Results for each participant across all conditions are presented in Figure 1. Missing data indicate a student’s absence on that day of the social skills group. The top panel of Figure 1 depicts Vito’s responding across baseline and the two video modeling conditions. Vito did not initiate with peers under baseline or the initial sharing conditions. Vito’s mean sharing immediately increased to 70% (range = 50%–100%) when video modeling was used to teach joining peers in an activity. During a reversal to the sharing condition, Vito demonstrated an immediate decrease in initiating with a mean of 11% (range = 0%–33%). Upon a final return to the joining condition, Vito demonstrated an increase in level ($M = 44\%$), although he did not show the same immediate increase in responding as had been observed during the first transition from sharing to joining. These results tentatively suggest video modeling was differentially effective as a function of the observed and direct stimuli associated with the different types of social initiations.

The middle panel of Figure 1 shows Reese’s target behaviors across all conditions. During baseline, Reese did not initiate with peers, resulting in a mean percentage of

0%. Reese’s initiations remained at 0% during the first sharing phase. A change in initiations was evident at the outset of the initial joining phase, during which Reese’s mean level of initiating was 78% (range = 25%–100%). When conditions were reversed to a second sharing phase, Reese’s initiating immediately returned to 0% and he demonstrated no instances of initiating during this phase. In a subsequent return to the joining condition, Reese’s initiations increased ($M = 33\%$), though not to the same levels as observed during the initial joining phase. Similar to Vito, these results tentatively suggest video modeling was differentially effective as a function of the observed and direct stimuli associated with the different types of social initiations.

The bottom panel of Figure 1 depicts Ivan’s target behaviors across all conditions. During baseline and the first sharing phase, Ivan did not engage in any initiations with peers. Although his mean performance during the first joining phase was 50% (range = 0%–100%), he demonstrated a consistent upward trend in percentage of initiations during this phase. During the return to sharing, Ivan did not engage in initiations with peers. In the final phase change, Ivan initiated with peers for 67% of trials across each of the three sessions. Given the limited number of data points in some phases for Ivan, these results are also somewhat tentative, but provide the strongest support of the three participants for a functional relation between the observed and direct stimuli associated with the joining condition and the performance of social initiations.

Percent of correct responding for first versus successive trials was also calculated for each participant to assess potential contribution from the video model versus the video model with corrective feedback. Vito, Reese, and Ivan had eight, seven, and eight opportunities, respectively, to respond to the first trial during the joining condition. Vito performed the target behavior on 62% of the first trial opportunities and 73% of the remaining trials in those sessions. During the sessions following an unsuccessful first response, Vito performed the target response for 45% of the remaining trials. Reese performed the target behavior on 57% of first trial opportunities and for 87% of the successive trials during those sessions. During the sessions with an error on the first trial, Reese performed the correct response on 43% of remaining trials. Ivan performed the correct response on the first trial for 25% of the sessions and for 75% of the remaining trials within those sessions. During the sessions where Ivan made an error on the first trial, he performed the behavior for 82% of the remaining trials. These outcomes suggest influence from both the video model and the correction sequence delivered following an error on the preceding trial.

Discussion

None of the participants engaged in the target behavior, initiating interactions with peers, during a baseline condition

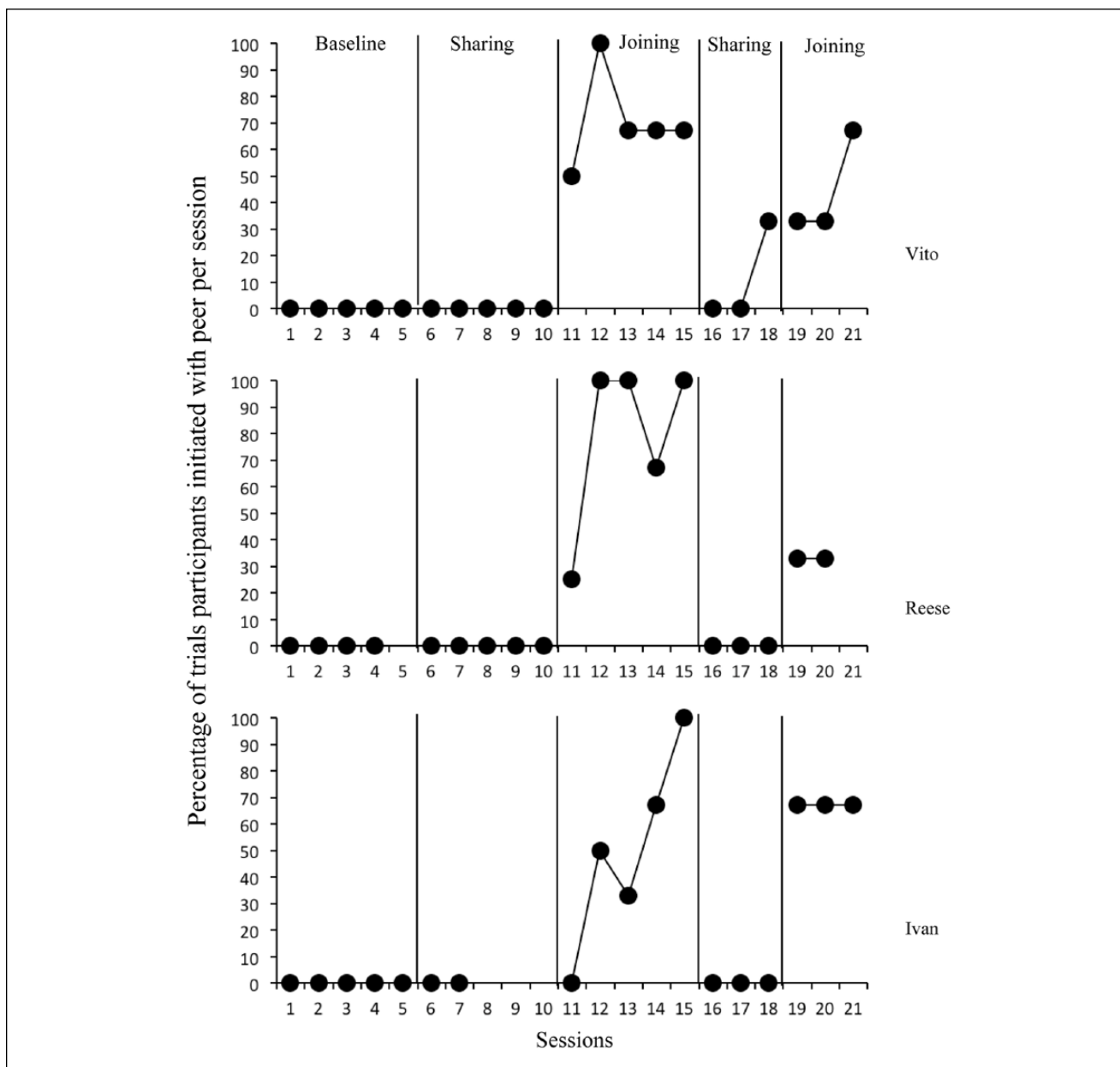


Figure 1. Percentage of trials each participant directed an initiation toward a peer during each of the three conditions: baseline, sharing, and joining.

and did not acquire the target behavior during the sharing condition. Subsequently, during the first phase of joining in play, all participants started directing initiations toward peers. Immediately upon reversal to a second sharing condition, all participants showed a reduction of initiating behavior to zero levels, where responding stabilized for all participants except Vito. Vito demonstrated a single instance of an initiation during the third session in this phase. Outcomes of joining in play were replicated to some extent when this condition was reinstated during the final phase of the experiment, though initiations did not completely return

to the levels established during the first joining phase. These outcomes show preliminary evidence of an association between environmental events depicted within video models, the corresponding environmental events experienced by the child, and the likelihood of participants to perform the observed behavior.

In addition to comparing responding across conditions, we analyzed the levels of responding during the first trial and subsequent trials within the joining condition. Responding to the first trial within a session speaks to the control of the video model over behavior, whereas responding after an error

suggests possible influence of the error correction sequence. As Vito and Reese performed the target behavior across greater than 50% of first trials, the video model appeared to influence behavior to some extent. During other sessions for Vito and Reese, and for the majority of sessions for Ivan, it is likely that the corrective feedback on a preceding trial also contributed to accurate performance on successive trials. This speaks to the importance of differential reinforcement delivered to the participant when preferred consequences are embedded within the video model.

Although preliminary, these results combine with previous research (i.e., Plavnick & Ferreri, 2011; Tetreault & Lerman, 2010) to inform methods that can increase the effectiveness of video modeling for children with autism. Previous research outside of video modeling has shown that children with autism whose responses contact reinforcement during instruction have more motivation to participate and respond correctly during future instruction (R. L. Koegel & Egel, 1979; L. K. Koegel et al., 2010). In addition, it has been suggested that observers are more likely to follow a successful as opposed to an unsuccessful model (Bandura, 1977). We hypothesize that participants in the present investigation were highly motivated to respond during the joining condition (see below) and that this motivation decreased during the sharing condition when participants observed the model receive a neutral or nonpreferred consequence. Given the efficiency of video modeling as an instructional procedure (Bellini & Akullian, 2007), it is important to ensure that components of the intervention, such as observed and direct consequences, maximize motivation to participate in learning.

Although several components were altered across the two conditions, the minimal responding during the sharing condition supports the theory that motivation influenced outcomes. As none of the participants performed the response in this condition prior to the final session of the second sharing phase, control over responding cannot be attributed to direct consequences (i.e., punishment) and is more likely connected to antecedent events. We anecdotally observed that participants engaged in few physical approaches toward peers during the sharing condition. Although tentative, as data on physical attempts were not formally collected throughout the study, fewer attempts during the sharing condition might suggest lower levels of motivation to perform the response.

Based on the above observations, we started collecting separate data on approach behavior during the final joining phase and noted that all participants engaged in the physical approach response on 100% of trials. However, an instance of correct responding required both physical and vocal components, and the vocal response did not occur as often as during the previous joining phase. Consistent attempts with a high rate of errors support the motivation hypothesis and also suggest additional instruction was needed for students

to consistently respond accurately. Given participants' limited level of language, the reversal back to the sharing condition may have weakened the probability that the correct vocal response would be selected and emitted during the final joining phase.

The results of the present study are contrary to some of the previous hypotheses regarding variability in the effectiveness of video modeling. First, negative video modeling outcomes could not be explained by child-specific characteristics or deficits in prerequisite skills (Delano, 2007; Nikopoulos & Keenan, 2003; Sherer et al., 2001) as these variables were constant across conditions. Second, the female models were also held constant across conditions indicating that gender matching of the model and observer may not be a necessary component for effective video modeling, despite suggestions that the model should be as similar to the observer as possible (Bellini & Akullian, 2007). The latter finding should be taken with caution as it is not known from the present investigation whether gender matching would be relevant for older individuals or if other characteristics, such as racial similarity, influence responding.

Although the present study demonstrates tentative support for the hypothesis that video modeling may be differentially efficacious based on the observed and direct consequences, there are several limitations that should be considered when analyzing the data and discussing the implications. First, the study was administered at the end of the public school year and there was limited time to complete the full reversal. As a result, the final sharing and joining phases were relatively brief. The brevity of these final phases limits the extent to which a full replication of results was possible (Kratochwill et al., 2013). Future research should attempt to replicate the present study by comparing the efficacy of video modeling for different target behaviors using an experimental design with at least five data points in each phase to demonstrate a clear functional relation (Kratochwill et al., 2013).

A second limitation was that the responses taught during the social skills group were not probed under other conditions. We noted that the children did not engage in initiations toward peers during our observations prior to the investigation. However, we were not able to conduct similar observations following exposure to the intervention and it is therefore unknown whether participants approached peers in natural environments after exposure to the intervention. The intervention period was relatively brief and generalization of the approach behavior was beyond the scope of this investigation. However, given previous reports of video modeling promoting generalization (e.g., Charlop-Christy, Le, & Freeman, 2000), future research should continue to investigate the generalization of skills involving peer interactions following video modeling interventions.

Third, the conditions were introduced to all three participants in the same order. As a result, it is possible that behavior during the joining condition was strengthened by the prior administration of sharing. These potential sequence effects appear unlikely given an almost full reversal to the absence of responding during the second sharing phase, though future research should counterbalance the order of conditions across participants.

Finally, although observed and direct consequences were manipulated across conditions, the video modeling procedures were administered as part of an instructional package that included video modeling, teacher instruction, naturalistic reinforcement, and error correction. It is impossible to know whether video modeling with embedded preferred consequences would be effective without including error correction sequences. In addition, it is impossible to separate the effects of the observed from direct consequences on participant performance.

The present investigation provides preliminary data showing that the effectiveness of video modeling may sometimes vary as a function of the target behavior and the environmental events associated with that behavior. This may be an important consideration when trying to understand outcome variability in previous video modeling research and in identifying the mechanisms responsible for positive effects within future video modeling research. From a practical perspective, it may be beneficial for service providers to carefully select initial target behaviors with an emphasis on skills that produce preferred consequences for the participant and to create video models depicting the relation between the selected behavior and preferred consequences.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

References

- American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed., text rev.). Washington, DC: Author.
- Bandura, A. (1965). Influence of models' reinforcement contingencies on the acquisition of imitative responses. *Journal of Personality and Social Psychology*, 1, 589–895.
- Bandura, A. (1977). *Social learning theory*. Englewood Cliffs, NJ: Prentice Hall.
- Bellini, S., & Akullian, J. (2007). A meta-analysis of video modeling and video self-modeling interventions for children and adolescents with autism spectrum disorders. *Exceptional Children*, 73, 264–287. Retrieved from <http://cec.metapress.com/content/122146>
- Cardon, T., & Azuma, T. (2012). Visual attending preferences in children with autism spectrum disorders: A comparison between live and video presentation modes. *Research in Autism Spectrum Disorders*, 6, 1061–1067. doi:10.1016/j.rasd.2012.01.007
- Charlop, M. H., & Milstein, J. P. (1989). Teaching autistic children conversational speech using video modeling. *Journal of Applied Behavior Analysis*, 22, 275–285. doi: 10.1901/jaba.1989.22–275
- Charlop-Christy, M. H., & Daneshvar, S. (2003). Using video modeling to teach perspective taking to children with autism. *Journal of Positive Behavior Interventions*, 5, 12–21. doi: 10.1177/10983007030050010101
- Charlop-Christy, M. H., Le, L., & Freeman, A. (2000). A comparison of video modeling with in vivo modeling for teaching children with autism. *Journal of Autism and Developmental Disorders*, 30, 537–552. doi:10.1023/A:1005635326276
- Delano, M. (2007). Video modeling interventions for individuals with autism. *Remedial and Special Education*, 28, 33–42.
- Gena, A., Couloura, S., & Kymissis, E. (2005). Modifying the affective behavior of preschoolers with autism using in-vivo or video modeling and reinforcement contingencies. *Journal of Autism and Developmental Disorders*, 35, 545–556. doi: 10.1007/s10803-005-0014-9
- Gilliam, J. (2006). *Gilliam Autism Rating Scale, 2nd Edition (GARS-2)*. North Tonawanda, NY: Multi-Health Systems.
- Koegel, L. K., Singh, A. K., & Koegel, R. L. (2010). Improving motivation for academics in children with autism. *Journal of Autism and Developmental Disorders*, 40, 1057–1066. doi: 10.1007/s10803-010-0962-6
- Koegel, R. L., & Egel, A. L. (1979). Motivating autistic children. *Journal of Abnormal Psychology*, 88, 418–426. doi: 10.1037/0021-843X.88.4.418
- Kratochwill, T. R., Hitchcock, J. H., Horner, R. H., Levin, J. R., Odom, S. L., Rindskopf, D. M., & Shadish, W. R. (2013). Single-case intervention research design standards. *Remedial and Special Education*, 34(1), 26–38. doi:10.1177/0741932512452794
- MacDonald, R., Sacramone, S., Mansfield, R., Wiltz, K., & Ahearn, W. H. (2009). Using video modeling to teach reciprocal pretend play to children with autism. *Journal of Applied Behavior Analysis*, 42, 43–55. doi: 10.1901/jaba.2009.42-43
- Mason, R. A., Ganz, J. B., Parker, R. I., Burke, M. D., & Camargo, S. P. (2012). Moderating factors of video-modeling with other as model: A meta-analysis of single-case studies. *Research in Developmental Disabilities*, 33, 1076–1086. doi: 10.1016/j.ridd.2012.01.016
- McGee, G. C., Almeida, M. C., Sulzer-Azaroff, B., & Feldman, R. S. (1992). Promoting reciprocal interactions via peer incidental teaching. *Journal of Applied Behavior Analysis*, 25, 117–126. doi: 10.1901/jaba.1992.25-117
- Miller, N. E., & Dollard, J. (1941). *Social learning and imitation*. New Haven, CT: Yale University Press.
- Nikopoulos, C. K., & Keenan, M. (2003). Promoting social initiation in children with autism using video modeling. *Behavioral Interventions*, 18, 87–108. doi: 10.1002/bin.129
- Nikopoulos, C. K., & Keenan, M. (2007). Using video modeling to teach complex social sequences to children with

- autism. *Journal of Autism and Developmental Disorders*, 37, 678–693. doi: 10.1007/s10803-006-0195-x
- Pace, G. M., Ivancic, M. T., Edwards, G. L., Iwata, B. A., & Page, T. J. (1985). Assessment of stimulus preference and reinforcer value with profoundly retarded individuals. *Journal of Applied Behavior Analysis*, 18, 249–255. doi: 10.1901/jaba/1985.18-249
- Plavnick, J. B., & Ferreri, S. J. (2011). Establishing verbal repertoires in children with autism using function-based video modeling. *Journal of Applied Behavior Analysis*, 44, 747–766. doi: 10.1901/jaba.2011.44-747
- Schopler, E., Van Bourgondien, M. E., Wellman, G. J., & Love, S. R. (2010). *Childhood Autism Rating Scale* (2nd ed.). Los Angeles, CA: Western Psychological Services.
- Sherer, M., Pierce, K. L., Paredes, S., Kisacky, K. L., Ingersoll, B., & Schreibman, L. (2001). Enhancing conversation skills in children with autism via video technology: Which is better, “self” or “other” as a model? *Behavior Modification*, 25, 140–158. doi: 10.1177/0145445501251008
- Taylor, B. A., Levin, L., & Jasper, S. (1999). Increasing play-related statements in children with autism toward their siblings: Effects of video modeling. *Journal of Developmental and Physical Disabilities*, 11, 253–264. doi: 10.1023/A:1021800716392
- Tetreault, A. S., & Lerman, D. C. (2010). Teaching social skills to children with autism using point-of-view video modeling. *Education & Treatment of Children*, 33, 395–419.

Copyright of Journal of Positive Behavior Interventions is the property of Sage Publications Inc. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.